



TRAINING PROGRAMME

AWS Serverless & Automation Projects

Six hands-on AWS projects, combined into one portfolio architecture

18 HOURS

FOR Engineers with AWS architecture basics who want hands-on portfolio projects



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Overview

This programme is built from projects. Participants deploy a scalable web application with Terraform, ship containers to ECS Fargate through GitHub Actions without long-lived keys, build a serverless API and an event-driven image processing pipeline, automate incident response, and add security gates to a container pipeline. Each project starts from a provided starter repository and ends with a working, torn-down deployment. The programme closes with a final project in which participants combine the projects into one documented portfolio architecture with infrastructure as code and a cost review.

At a glance

#	PHASE	FOCUS	HOURS
01	Containers & Infrastructure as Code	Terraform web stack, CI/CD to ECS Fargate with OIDC	6 h
02	Serverless & Security Automation	Serverless API, event-driven pipeline, incident automation, image scanning gates	6 h
03	Final Project	One documented portfolio architecture	3 h
	Buffer	Catch-up, revision, extra lab time	3 h
	Total		18 h

Key facts

AUDIENCE	Engineers with AWS architecture basics who want hands-on portfolio projects
PREREQUISITES	AWS architecture (course: AWS Solutions Architect Associate); Terraform and GitHub Actions basics (course: DevOps Engineering). These are used, not taught
DURATION	18 h: 15 h of chapters + 3 h buffer
FORMAT	Project labs from provided starter repositories, each with a short design briefing
LAB ENVIRONMENT	One laptop per participant (Terraform, AWS CLI, Git, Docker); a GitHub repository per participant and an AWS sandbox account per team, provided by the organiser, with a budget alert; every project destroyed with terraform destroy at the end of the lab to control cost
ASSESSMENT	Each project validated when it runs, and a final portfolio project presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants leave with six working AWS projects in Git, combined into one documented architecture they can explain, rebuild and cost

PHASE 01

6 H

Containers & Infrastructure as Code

Participants provision a scalable web stack with Terraform from a starter repository, then deliver a containerised version through GitHub Actions to ECS Fargate, authenticating to AWS with OIDC instead of stored keys. Validated projects: the web stack answering behind its load balancer, and a commit that reaches Fargate through the pipeline.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Scalable Web App	Terraform: VPC, Application Load Balancer, Auto Scaling, RDS, from a provided starter repository	Lab	3 h
1.2	CI/CD to ECS Fargate	GitHub Actions with OIDC, Amazon ECR, Amazon ECS on Fargate	Lab	3 h

PHASE 02

6 H

Serverless & Security Automation

Participants build serverless and event-driven systems: an API on Lambda, API Gateway and DynamoDB, an image processing pipeline triggered by S3 events, automated incident response from GuardDuty and CloudWatch through EventBridge, and a container pipeline that stops on critical vulnerabilities. Validated projects: each of the four running and demonstrated with a test event.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Serverless API	Lambda, API Gateway, DynamoDB	Lab	1.5 h
2.2	Image Processing Pipeline	S3 events, Lambda, notifications	Lab	1.5 h
2.3	Incident Response Automation	GuardDuty and CloudWatch to EventBridge to Lambda and SNS	Lab	1.5 h
2.4	Container Security Pipeline	ECR scanning, Amazon Inspector, build gates	Lab	1.5 h

PHASE 03

3 H

Final Project

Each participant combines the six projects into one portfolio architecture, documented so that another engineer could rebuild it, and presents it to the group.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Integrate & Document	Combine the projects in one repository with infrastructure as code, architecture diagram and README	Project	1.5 h
3.2	Cost Review, Demo & Teardown	Cost review, live demo, full teardown; questions and feedback	Project	1.5 h

FINAL PROJECT

AWS Portfolio Architecture

The six projects combined into one documented architecture: a web tier and containers on Fargate, a serverless API and event pipeline, incident automation and image scanning gates, all rebuilt from code and reviewed for cost.

STACK AWS: VPC, ALB, Auto Scaling, RDS, ECR, ECS Fargate, Lambda, API Gateway, DynamoDB, S3, EventBridge, SNS, GuardDuty, Inspector; Terraform, GitHub Actions

DELIVERABLE Git repository with IaC and pipelines, architecture diagram, cost review, live demo